PROBABILITY AND STATISTICS CASE STUDY

TITLE PAGE

Course: BAMAT207 Probability and Statistics

Project title: Statistical Modelling of Packet-Loss Patterns in Computer Networks

Focus: Distribution fitting, tests of independence and Bayesian estimation

Tracks: Track 3 (core), Track 1 and Track 5 (supporting)

Stage: Final review (Phase 1 simulation study, Phase 2 real-network lab)

TEAM

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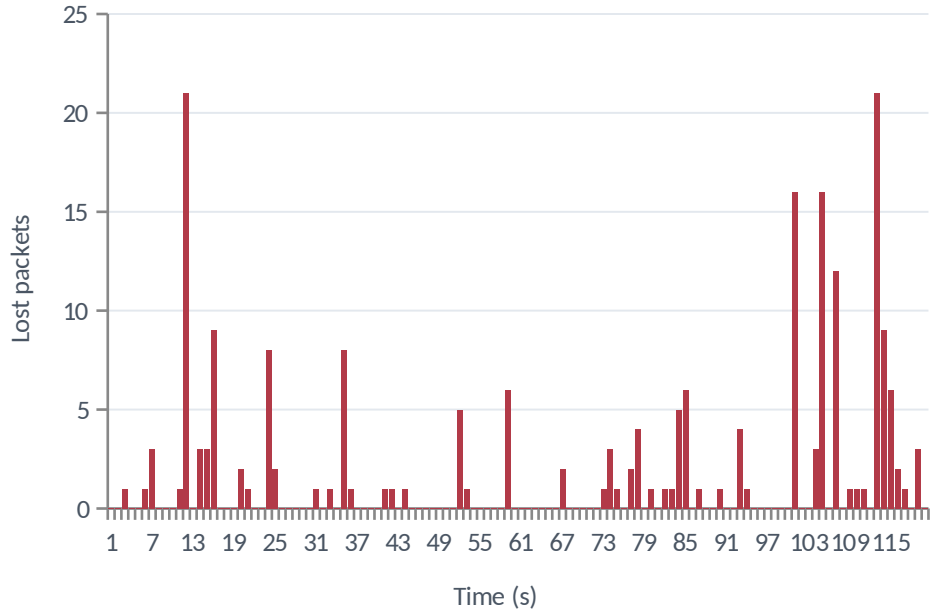
Aashman Biyani
25BBS0100

3
network conditions

540,000
packets analysed

6
objectives

SAMPLE TRACE



Each bar is one second of traffic. Several losses often fall in the same second.

PROGRESS SINCE REVIEW 1

SHOWN AT REVIEW 1

Problem: objectives and hypotheses.

Model: Good/Bad loss model and simulation design.

Results: distribution fits and independence tests.

Inference: Bayesian estimates and interval coverage.

Planning: power study for Phase 2.

ADDED SINCE

Testing: automated tests for edge cases, correctness and reproducibility.

Fixes: empty traces and zero-loss cases now handled safely.

Phase 2: real-packet probe and the Real Network Lab.

Web app: Packet-Loss Lab with live and simulated pages.

IEEE paper: seven-page conference paper with all results.

FINAL DELIVERABLES

Code: one Python pipeline, public on GitHub.

Lab: streamlit run lab.py records and analyses real runs.

Results: 7 figures, results.json and summary.md.

Paper: IEEE format, PDF and Word.

Slides: this deck, built from the same data.

540,000
packets analysed

2,000
simulated captures per Monte Carlo point

2026
seed: every number reproducible

7 pages
IEEE conference paper

INTRODUCTION

Basic loss models assume that each packet is lost independently. Under this assumption, loss counts are Poisson and gaps are exponential. Measured traffic often shows losses grouped close together.

CONTEXT

REAL-TIME MEDIA
Several losses in a row are harder to conceal than isolated ones.

PROTOCOL SETTINGS
Retransmission and error-correction settings use loss statistics.

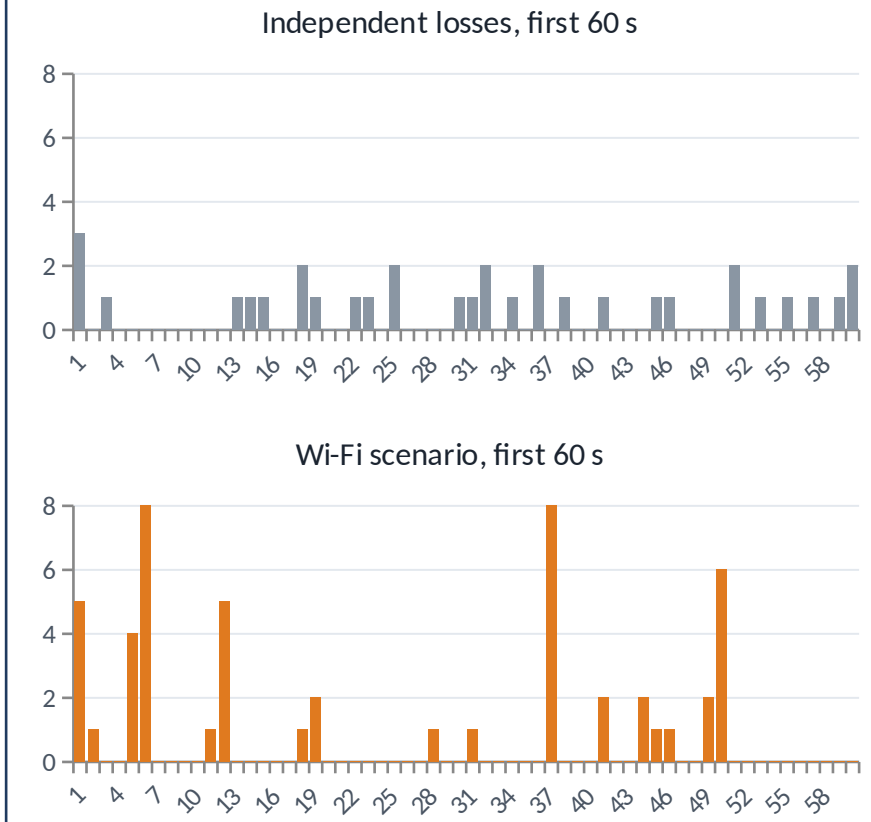
SERVICE LEVELS
A reported loss rate is only useful if the model behind it holds.

STATISTICAL INFERENCE
Ignoring dependence makes standard errors too small.

RESEARCH QUESTIONS

1. Do loss counts and gaps follow the Poisson model?
2. Which distributions describe them better?
3. How much does dependence affect interval estimates?

INDEPENDENT AND DEPENDENT LOSSES



Both series have a loss rate of 1.29%. The Wi-Fi series has more seconds with no loss and a few seconds with many.

OBJECTIVES AND HYPOTHESES

OBJECTIVES

O1 LOSS COUNTS
Compare Poisson and Negative Binomial models for losses per second.

O2 INTER-LOSS GAPS
Fit exponential and two-phase hyperexponential models.

O3 INDEPENDENCE
Runs test and Markov chain likelihood ratio test.

O4 NETWORK COMPARISON
Compare the 3 networks with rank-based tests.

O5 BAYESIAN ESTIMATION
Loss rates and burst parameters with credible intervals.

O6 RELIABILITY
Power of the tests and coverage of the intervals.

HYPOTHESES

	Null hypothesis H0	Alternative H1
H1	Losses per second are Poisson (random)	Too spread out: losses cluster
H2	Gaps between losses are exponential	Gaps are not exponential
H3	Each packet is lost independently	A loss changes the next packet's chance: $P(L\|L) \neq P(L\|N)$
H4	All networks behave the same	At least one network differs

All tests use a significance level of $\alpha = 0.05$.

TRACK MAPPING

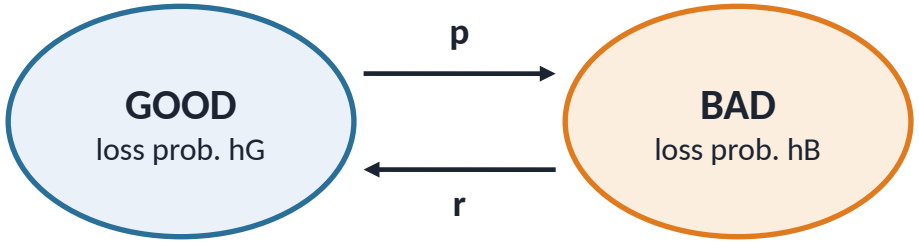
Track 3 (core):
O1 and O2, distribution modelling

Track 5:
O3, O4 and O6a, hypothesis testing

Track 1:
O5 and O6b, Bayesian inference

LOSS MODEL AND DATA

GILBERT-ELLIOTT TWO-STATE MODEL



πB = p / (p + r)
Share of time the network spends in the Bad state.

P(loss) = (1 − πB) × hG + πB × hB
Total loss = Good loss × Good time + Bad loss × Bad time.

Average stay = 1/p in Good, 1/r in Bad
If you leave with chance r per packet, you stay about 1/r packets.

The simulation draws complete sojourn times from the geometric distribution instead of updating the state for every packet.

SCENARIO PARAMETERS

Network	p	r	hG	hB	πB	True loss	Mean Bad spell
Wired	0.0005	0.3	0.002	0.5	0.17%	0.283%	3.3 pkts
Wi-Fi	0.002	0.15	0.004	0.6	1.32%	1.184%	6.7 pkts
Mobile	0.004	0.08	0.006	0.7	4.76%	3.905%	12.5 pkts

p and r are per-packet transition probabilities. hG and hB are the loss probabilities in each state.

SAMPLING SETUP

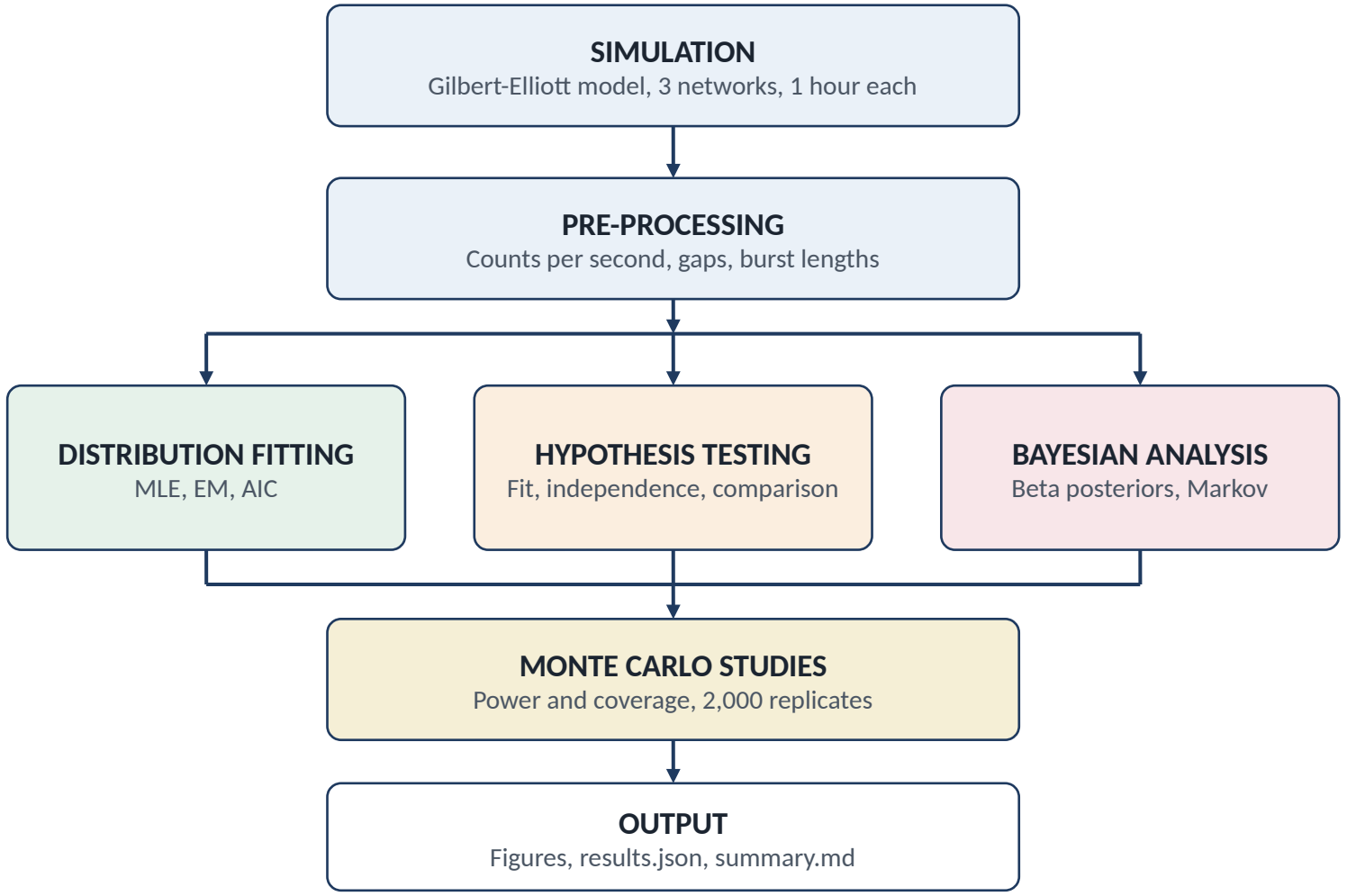
50/s	1 hour	180,000	2026
packets (20 ms apart)	per network	packets each	random seed

PHASE 1 (COMPLETED)
Simulation with known parameters.

PHASE 2 (PLANNED)
Measured traces, same code.

METHODOLOGY

ANALYSIS PIPELINE



REASON FOR A SIMULATION PHASE

- The true parameters are known in advance.
- Estimates can be compared with the true values.
- The power study gives the capture length for Phase 2.
- Phase 2 uses the same code on measured data.

SOFTWARE

Python 3 | **NumPy**

SciPy | **Matplotlib**

Each statistical component is a separate module with one owner.

STATISTICAL METHODS

METHODS BY OBJECTIVE

Obj.	Method	Formula	What it does
O1	Chi-square goodness of fit	$\chi^2 = \Sigma\ (O - E)^2 / E$	Checks if the model's prediction matches what really happened.
O1	Index of dispersion	$D = s^2 / \bar{X}$	Checks if losses are more spread out than random (clumping).
O2	Kolmogorov-Smirnov (bootstrap)	$D = \max\ \lvert Fn(t) - F(t) \rvert$	Checks if the waiting times between losses are random.
O2	EM algorithm	guess → update → repeat	Splits the waiting times into short (in a burst) and long ones.
O3	Runs test	$z = (R - E[R]) / sd(R)$	Checks if lost packets stick together in clumps.
O3	Markov likelihood ratio	$G^2 = 2\ (\ln L1 - \ln L0)$	Checks if one loss makes the next loss more likely.
O4	Kruskal-Wallis, Mann-Whitney	$p \times$ number of pairs	Checks if the networks are really different, not just by luck.
O5	Bayesian update	$Beta(1 + k, 1 + n - k)$	Estimates the loss rate and how sure we are.
O6	Monte Carlo	$SE = \sqrt{(p(1 - p) / N)}$	Repeats the experiment 2,000 times to check our methods work.

AIC = 2k − 2 ln L
Picks the better model: the lower score wins.

Count window:
1 s = 50 packets, 3,600 windows

Design-effect blocks:
10 s = 500 packets

DESCRIPTIVE STATISTICS

1 HOUR PER NETWORK, 180,000 PACKETS EACH

Network	Loss rate	Mean / s	Variance	Var ÷ mean	Mean burst	Max burst
Wired	0.27%	0.138	0.219	1.59	1.12	6
Wi-Fi	1.29%	0.646	3.430	5.31	1.55	9
Mobile	3.94%	1.969	20.017	10.17	2.25	16

A burst is a run of consecutive lost packets.

KEY FIGURES

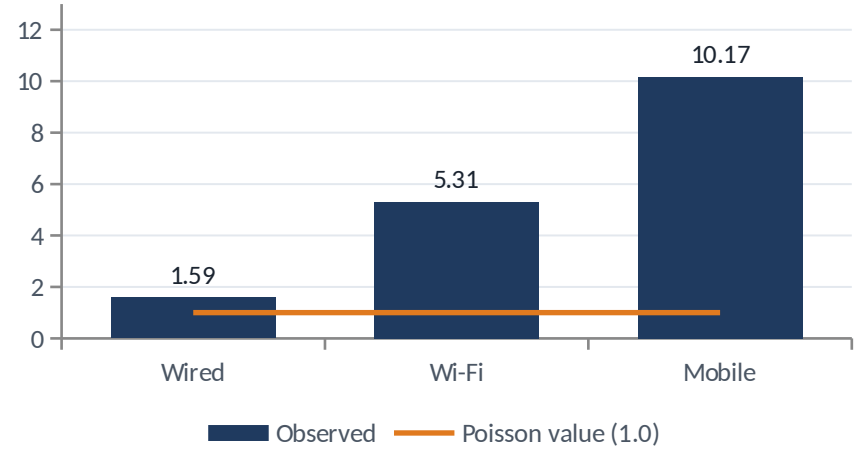
10.17
highest var ÷ mean (Mobile)

16
longest burst, packets (Mobile)

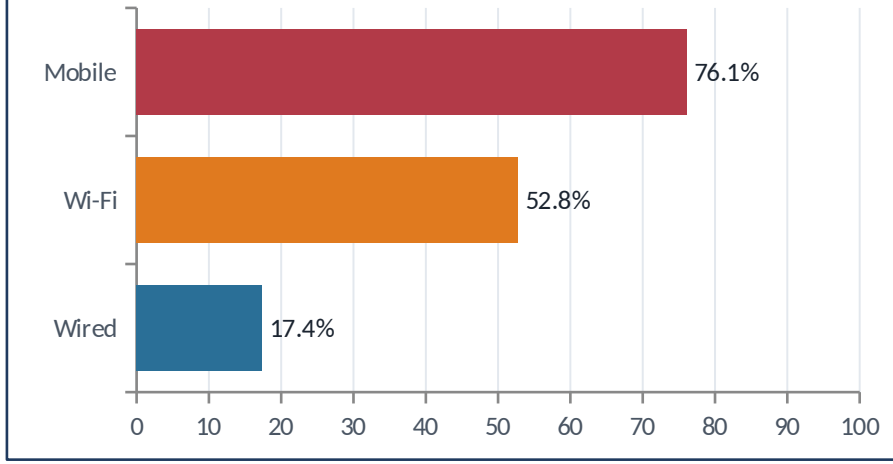
40
most losses in one second (Mobile)

For a Poisson process the variance-to-mean ratio would be close to 1. All 3 networks are above this value.

VARIANCE-TO-MEAN RATIO

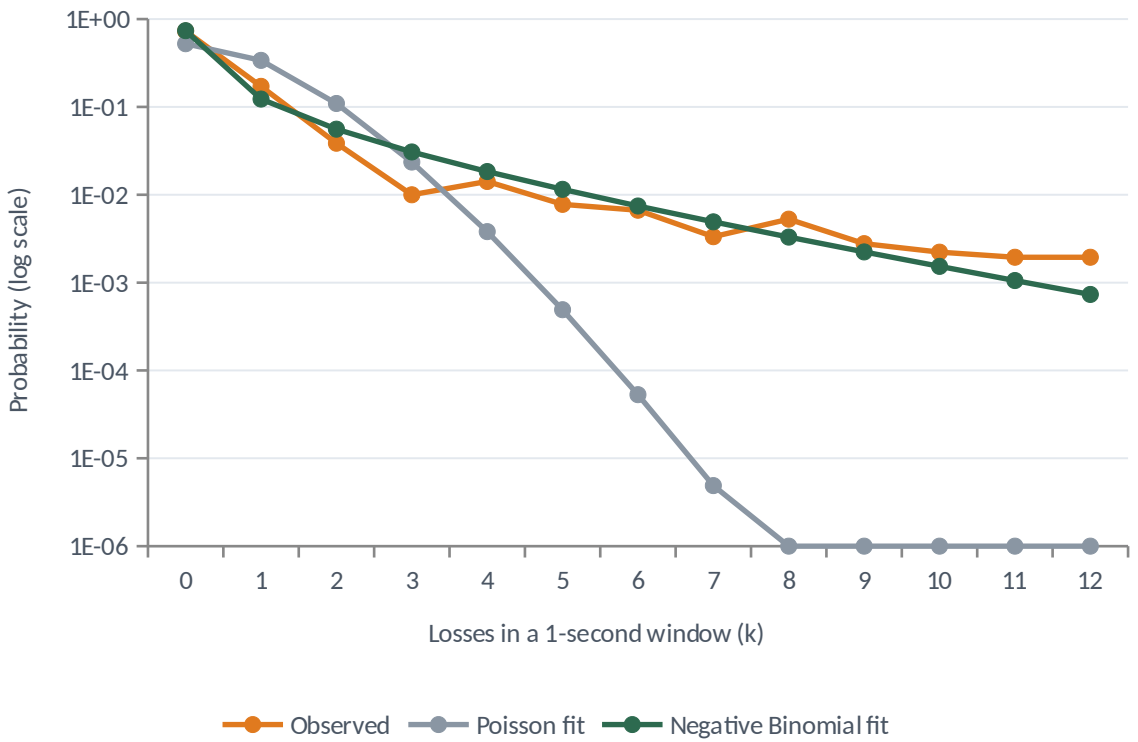


SHARE OF LOSSES INSIDE BURSTS



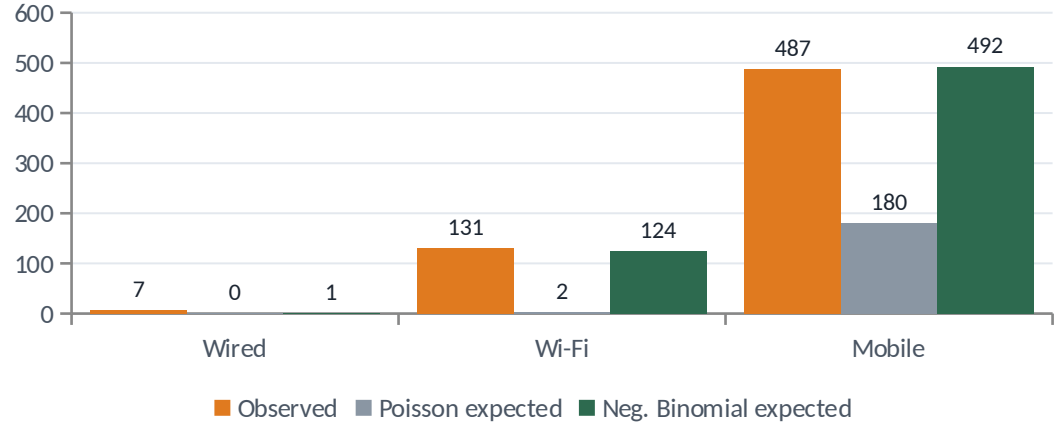
DISTRIBUTION OF LOSS COUNTS

WI-FI: PROBABILITY OF K LOSSES IN ONE SECOND



Values below 10^{-6} are plotted at the axis minimum.

SECONDS WITH 5 OR MORE LOSSES (OF 3,600)



GOODNESS OF FIT

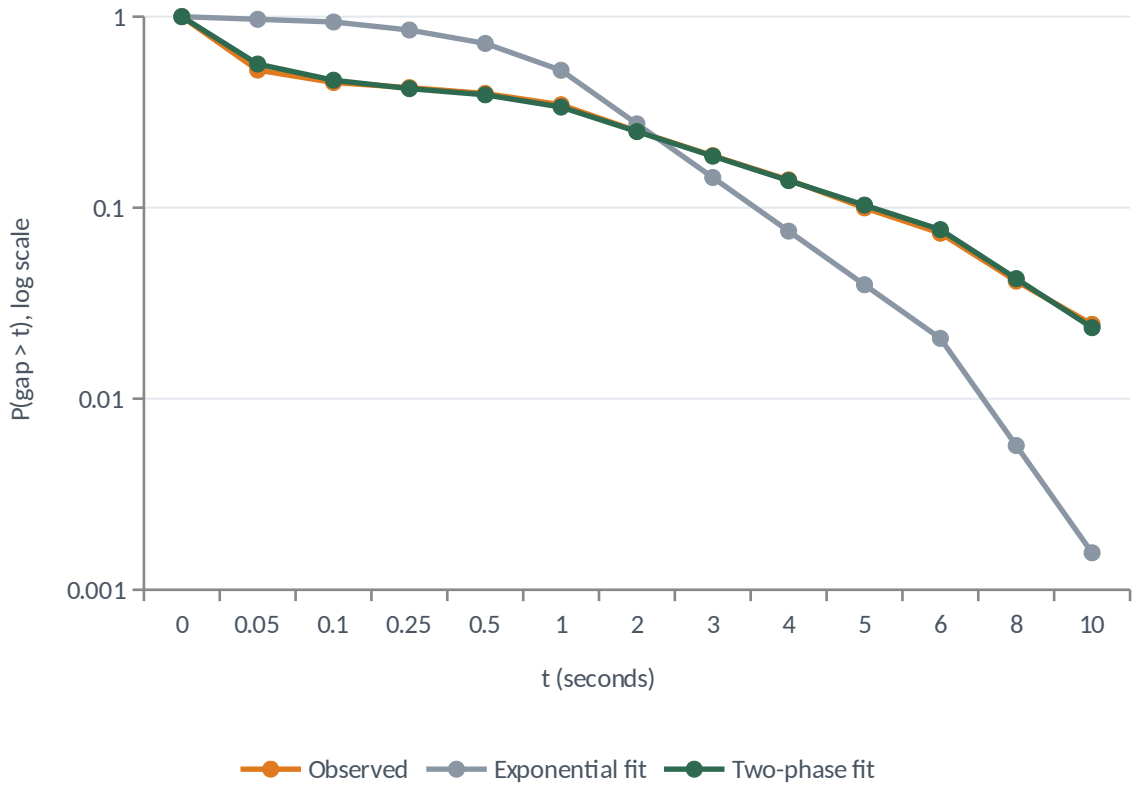
Network	Dispersion D	χ^2 Poisson	AIC Poisson	AIC Neg. Bin.
Wired	1.59	22.5	3,143	2,992
Wi-Fi	5.31	2544.2	10,607	7,186
Mobile	10.17	15840.8	25,042	11,700

RESULT: POISSON MODEL REJECTED FOR ALL 3 NETWORKS (DISPERSION TEST $p < 10^{-10}$)

The Negative Binomial is a Poisson distribution whose rate follows a Gamma distribution. This lets the variance exceed the mean. Its AIC is lower for every network.

DISTRIBUTION OF INTER-LOSS GAPS

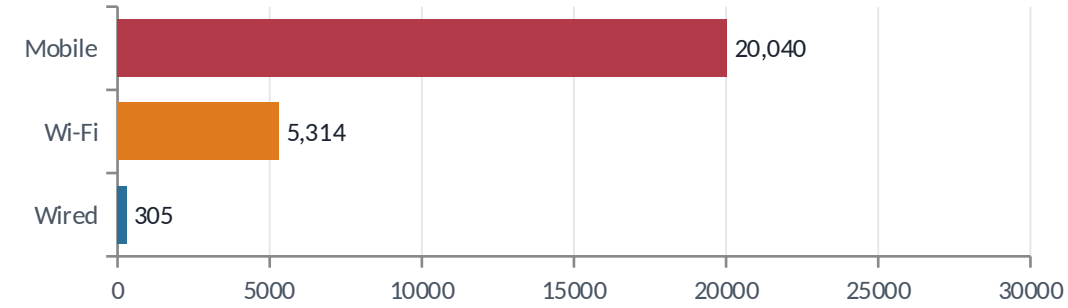
WI-FI: PROBABILITY THAT A GAP EXCEEDS T



TWO-PHASE HYPEREXPONENTIAL FIT (EM)

Network	Fast share	Fast mean	Slow mean	KS exp. p	KS mix. p
Wired	16%	32 ms	8.6 s	< 0.0005	0.15
Wi-Fi	55%	33 ms	3.4 s	< 0.0005	< 0.002
Mobile	77%	29 ms	2.1 s	< 0.0005	< 0.002

REDUCTION IN AIC WITH THE TWO-PHASE MODEL

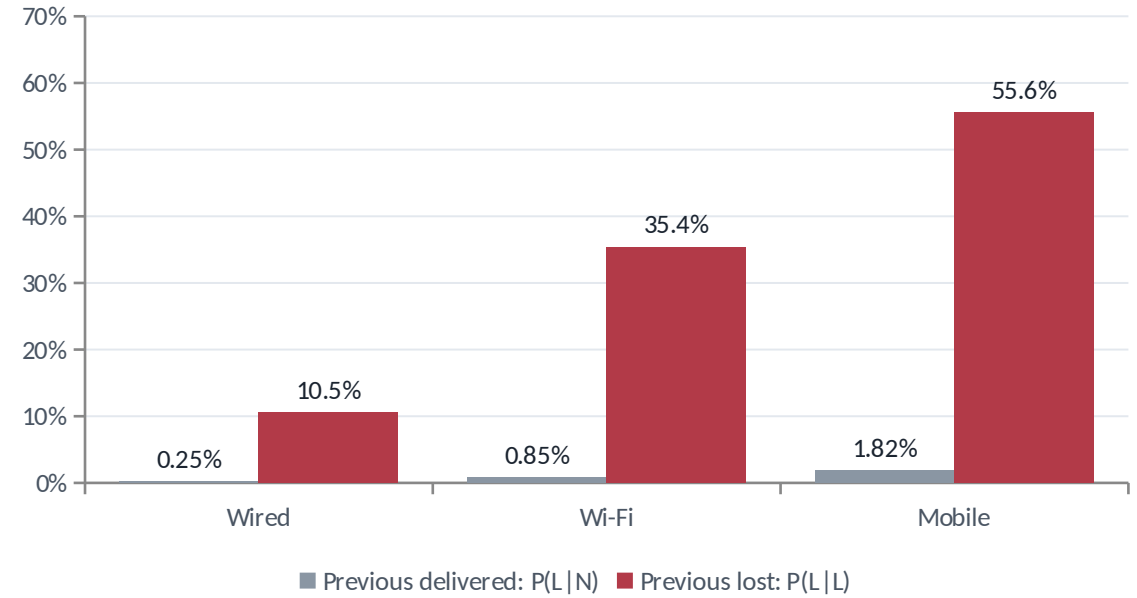


FAST PHASE: GAPS INSIDE A BURST
Mean gap between 29 and 33 ms, about one or two packet intervals.

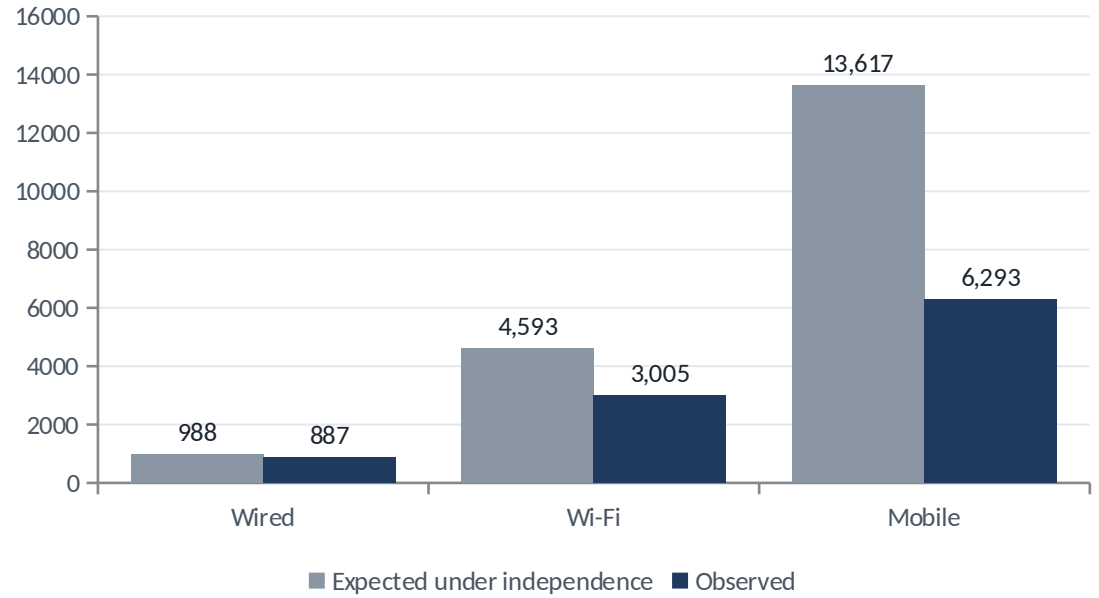
SLOW PHASE: GAPS BETWEEN BURSTS
Mean gap 8.6 s (Wired), 3.4 s (Wi-Fi), 2.1 s (Mobile).

TESTS OF INDEPENDENCE

LOSS PROBABILITY GIVEN THE PREVIOUS PACKET



NUMBER OF RUNS: EXPECTED AND OBSERVED



TEST RESULTS

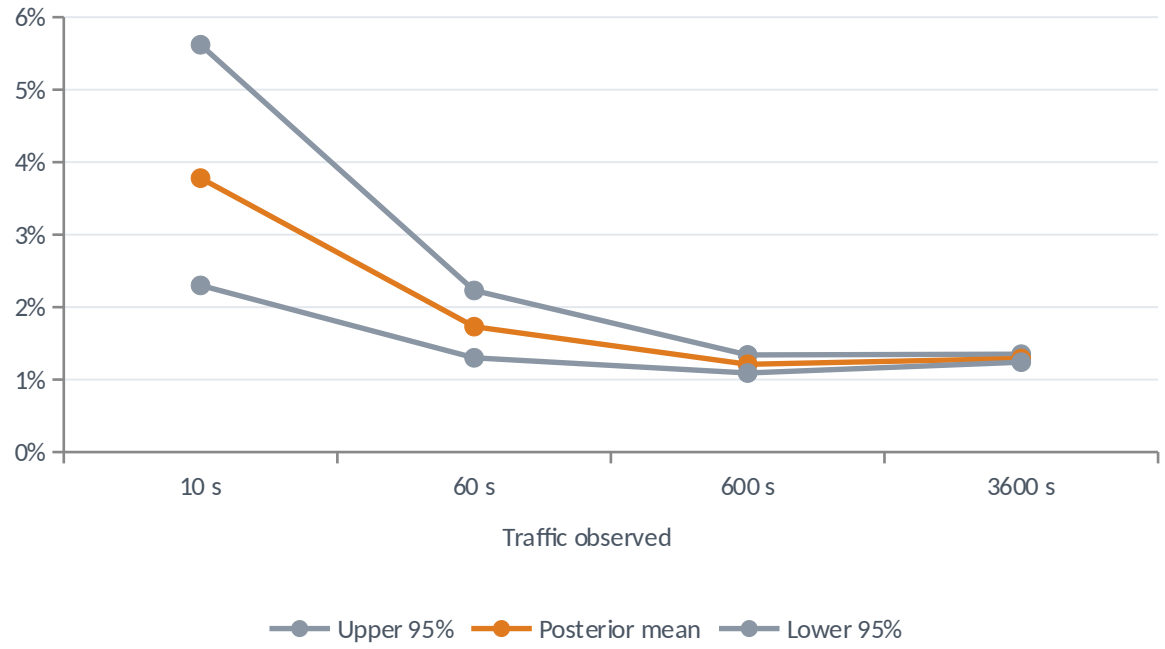
Network	Runs z	Markov G^2	p-value	Ratio b/a	Mean burst
Wired	-43.5	288.4	$< 10^{-10}$	43×	1.12 pkts
Wi-Fi	-146.7	4498.6	$< 10^{-10}$	42×	1.55 pkts
Mobile	-228.2	18560.2	$< 10^{-10}$	31×	2.25 pkts

31× to 43×

Increase in loss probability after a loss (posterior median of b/a). H3 is rejected for every network.

BAYESIAN ESTIMATION

WI-FI: POSTERIOR AFTER MORE TRAFFIC



After 10 s: 3.78% [2.30%, 5.62%]. After 3600 s: 1.29% [1.24%, 1.35%].

POSTERIOR LOSS RATE, FULL TRACE

Network	Posterior mean	95% credible interval
Wired	0.276%	[0.252%, 0.300%]
Wi-Fi	1.293%	[1.241%, 1.345%]
Mobile	3.938%	[3.848%, 4.028%]

COMPARING THE NETWORKS (H4)

Kruskal-Wallis: $H = 937.3$, $p < 10^{-10}$

Chi-square homogeneity: 7143.0 on 2 df

Mann-Whitney pairs: largest Bonferroni $p < 10^{-10}$

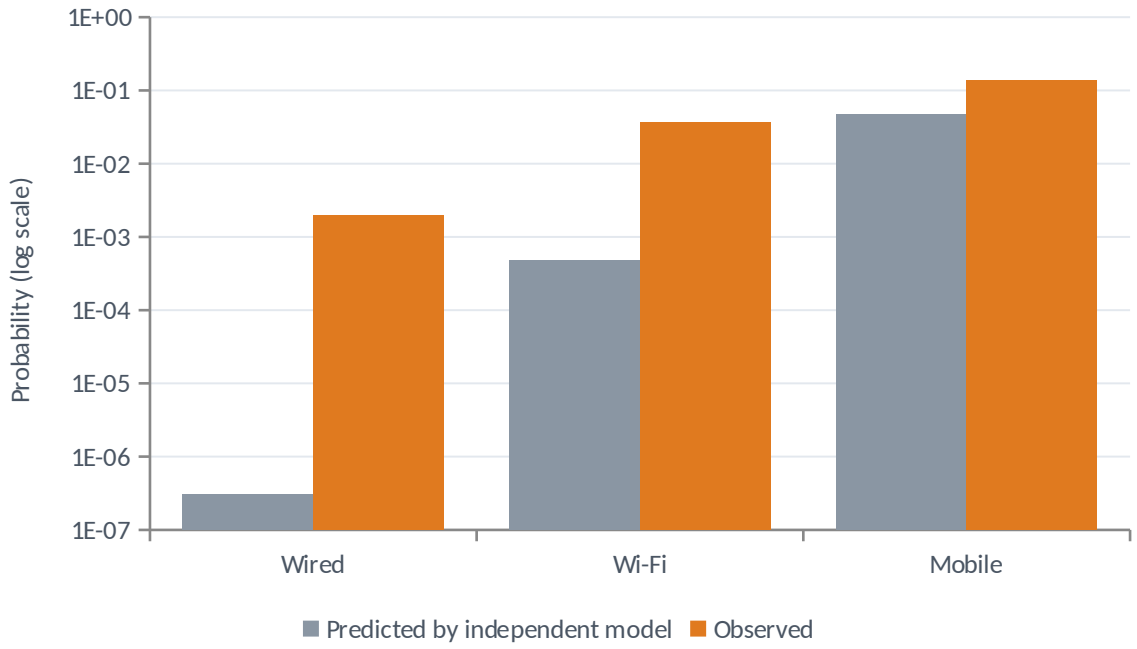
Beta(1, 1) → Beta(1 + k, 1 + n − k)
Start knowing nothing. Add the k losses and n − k deliveries you saw.

ORDERING
P(Wired < Wi-Fi < Mobile) = 1.000 from 200,000 posterior draws.

MEAN BURST LENGTH
Posterior medians of 1.12, 1.55 and 2.25 packets.

PREDICTIVE CHECK AND DESIGN EFFECT

P(5 OR MORE LOSSES IN ONE SECOND)



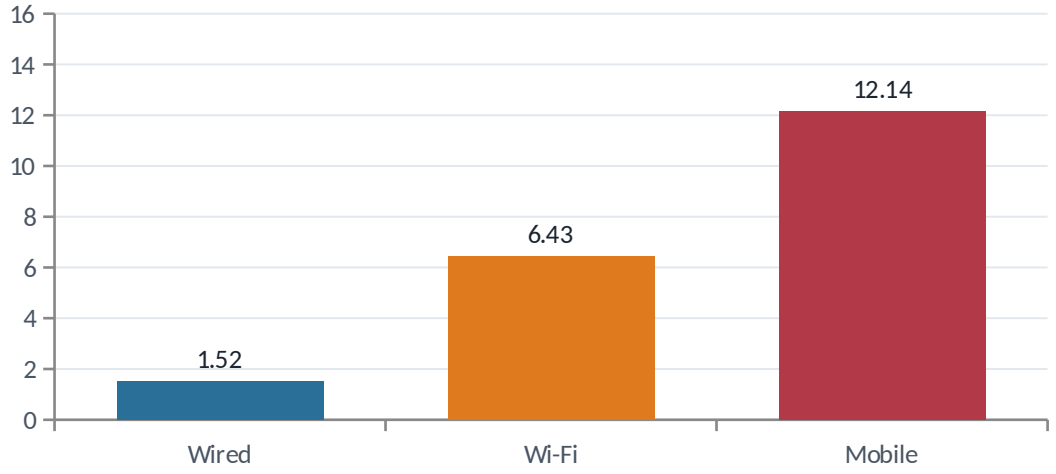
6,300×
Wired

77×
Wi-Fi

2.9×
Mobile

Observed frequency divided by the frequency predicted under independence.

DESIGN EFFECT (10-SECOND BLOCKS)



EFFECT OF POSITIVE CORRELATION

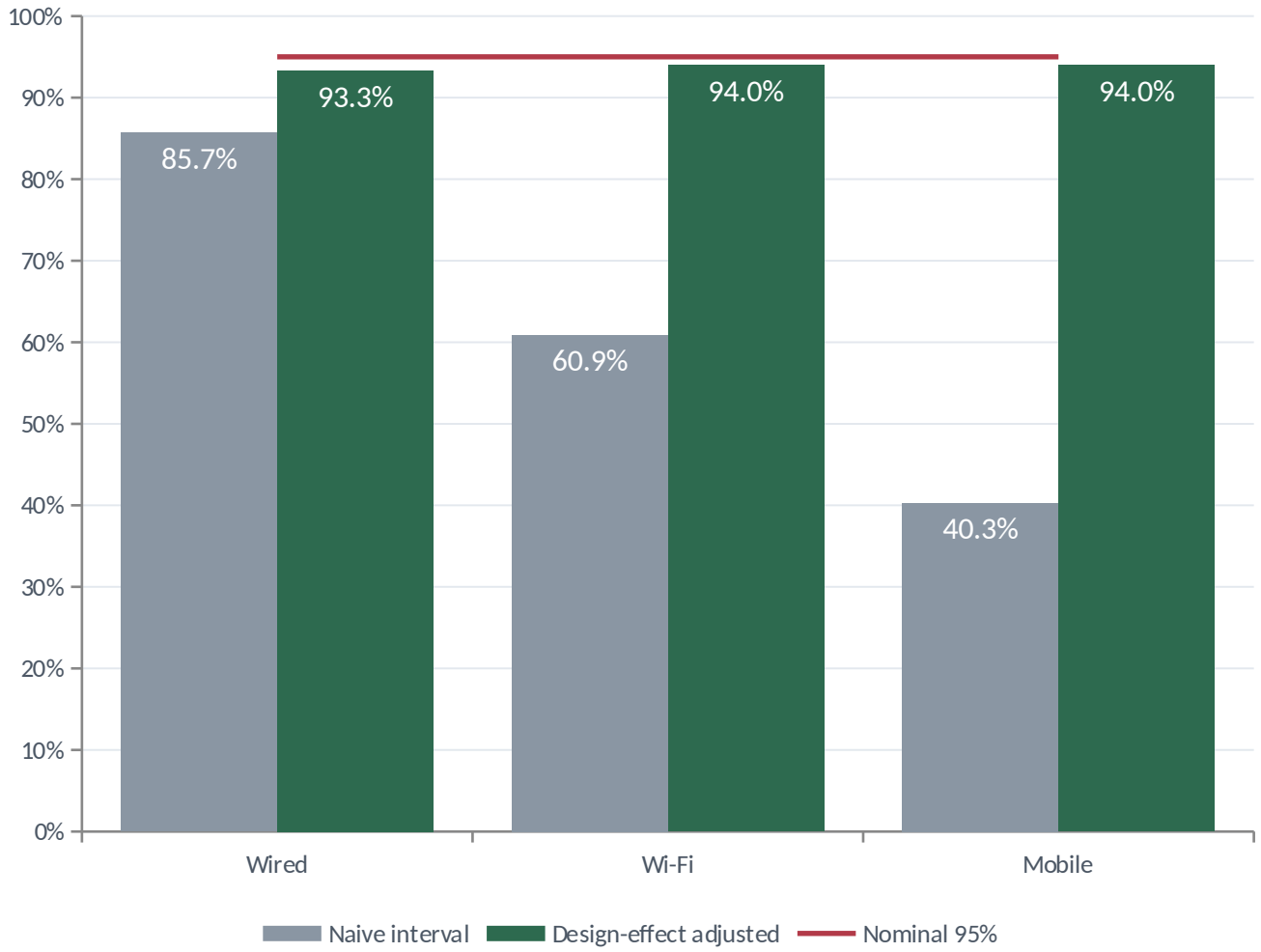
$$\mathrm{Var}(\Sigma X_i) = \Sigma \mathrm{Var}(X_i) + 2\Sigma \mathrm{Cov}(X_i, X_j)$$

Total spread = each packet's own spread + extra spread from packets moving together

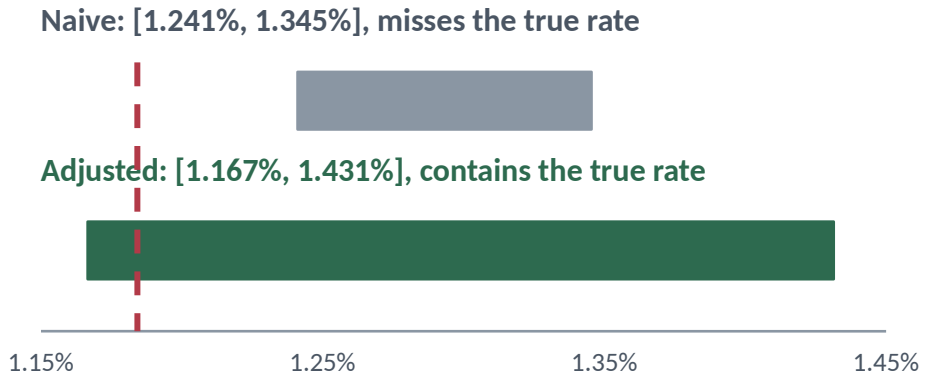
- Inside a burst, neighbouring packets are positively correlated.
- Design effect = real spread ÷ spread if losses were random.
- Effective sample size n ÷ deff: for Wi-Fi, 180,000 packets count as about 28,000.

COVERAGE OF 95% INTERVALS

COVERAGE IN 2,000 SIMULATED 10-MINUTE CAPTURES



EXAMPLE: WI-FI, FULL TRACE



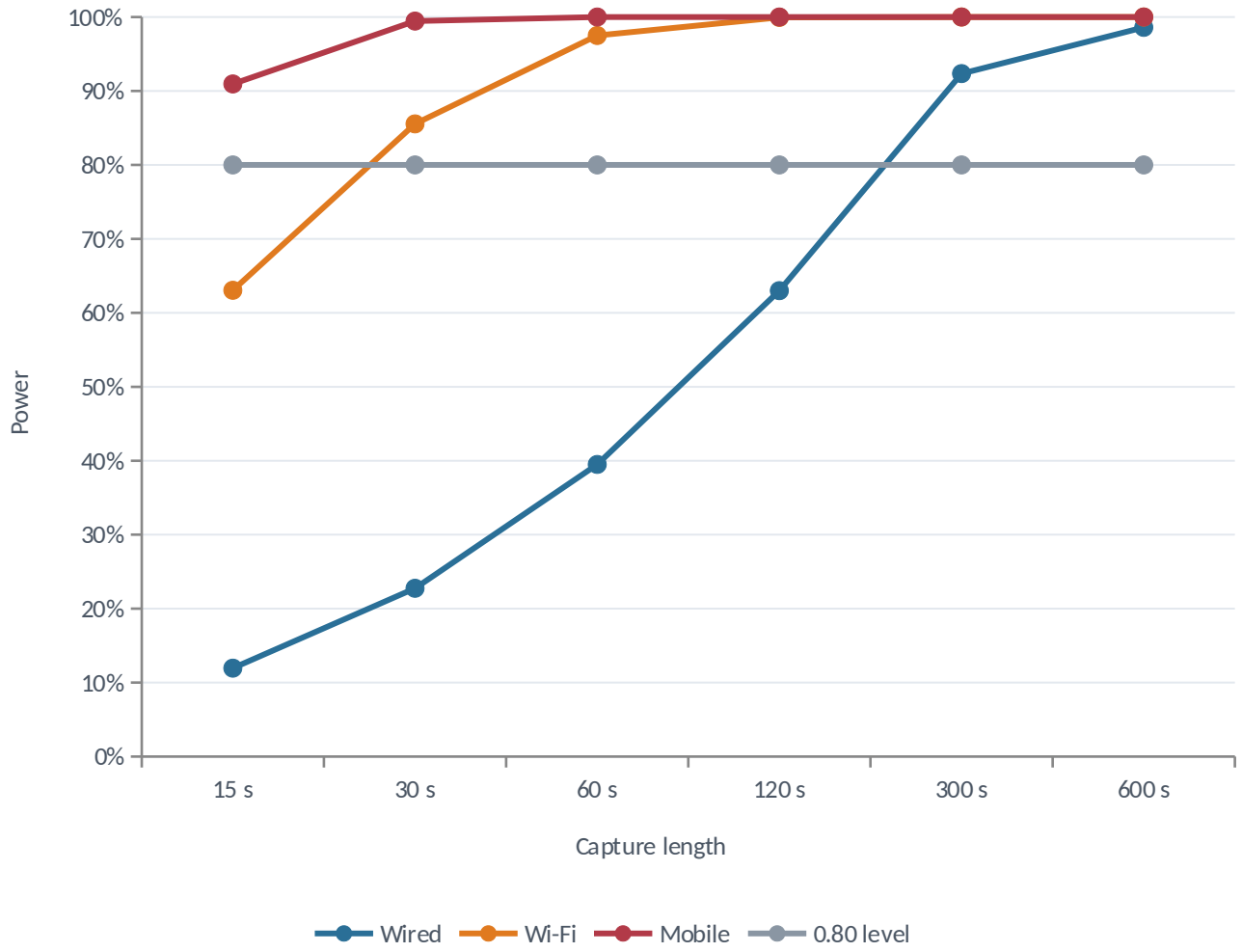
AVERAGE INTERVAL WIDTH

Network	Naive	Adjusted
Wired	0.120 pp	0.155 pp
Wi-Fi	0.245 pp	0.563 pp
Mobile	0.438 pp	1.599 pp

pp = percentage points.

POWER ANALYSIS

POWER OF THE RUNS TEST BY CAPTURE LENGTH



SHORTEST CAPTURE WITH POWER ≥ 0.80

Network	Dispersion	Runs	Markov
Wired	300 s	300 s	300 s
Wi-Fi	60 s	30 s	30 s
Mobile	15 s	15 s	15 s

The runs test needs the shortest capture for every network.

2,000
replicates per point

≤ 0.011
Monte Carlo standard error

RECOMMENDATION FOR PHASE 2
Record at least 10 minutes per network. At this length every test has power of at least 0.97.

PHASE 2: MEASURING REAL NETWORKS

HOW A REAL RUN WORKS

1. Send
50 numbered packets per second (one every 20 ms) to the router by ICMP ping, or by UDP to an echo server.

↓

2. Match
Each reply is matched to its packet by sequence number; the round-trip time is stored.

↓

3. Decide
No reply within 2 s means the packet is lost.

↓

4. Record
The run becomes a 0/1 sequence, saved as CSV with its settings.

↓

5. Analyse
The Phase 1 pipeline (O1 to O6) runs on it automatically.

VALIDATION: SAME LOSS RATE, BURSTY VS RANDOM

Measure	Test server: bursty drops	Test server: random drops
Loss rate	1.28% (77 of 6,000)	1.22% (73 of 6,000)
Variance ÷ mean, D	3.82	0.92
Runs test z (p)	−24.5 (< 10^{-10})	−1.2 (0.231)
b/a	36.8	3.0
Design effect	4.43	1.00
Verdict	Losses come in bursts	No evidence of bursts

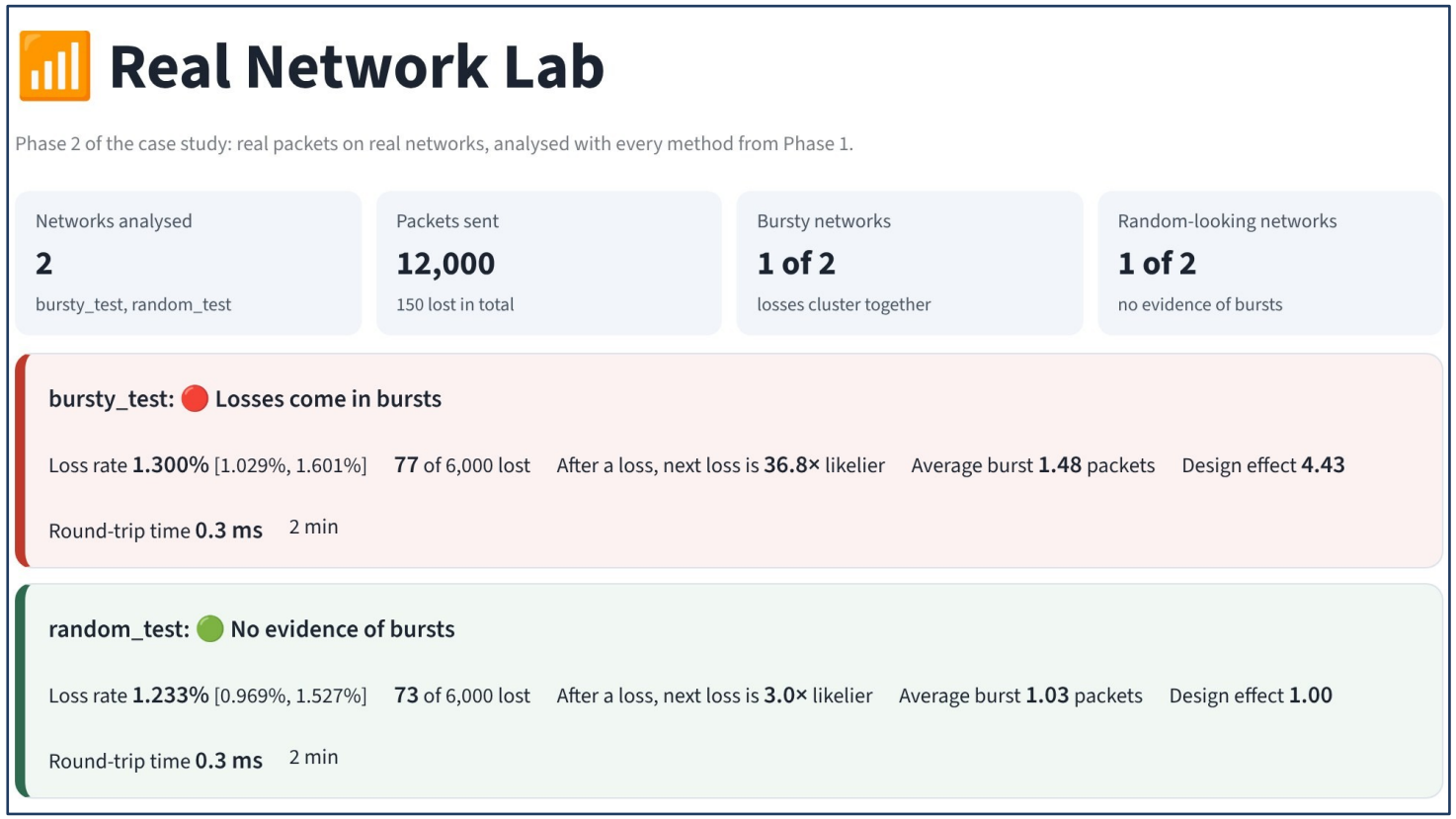
2-minute runs over the laptop's own loopback; the test server dropped replies on purpose.

50/s
packets, voice-call rate

2 s
timeout for a reply

10 minutes
per network: power ≥ 0.97

REAL NETWORK LAB: THE FINAL PRODUCT



Home page with the two validation runs: one card per network, then a tab per objective.

WHAT IT DOES BY ITSELF

Record: pick the network, press Start; live loss rate on screen.

Verdict cards: bursty or random, loss rate with 95% CrI, b/a, deff.

O1 to O6 tabs: every fit, test and interval from Phase 1, as charts.

Live vs study: each run set against the matching study network.

Assumptions: what we assumed, what we found, and why.

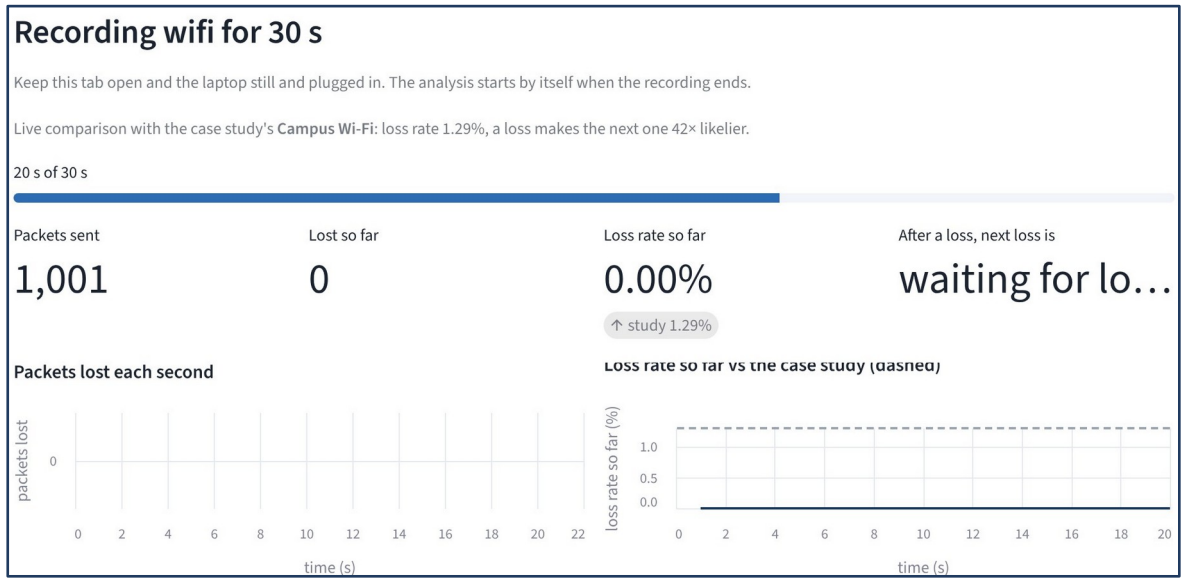
Saved runs: keep runs with a name and note; compare all.

Read me: every formula and decision rule in the sidebar.

RUN IT
streamlit run lab.py (from the project folder)

REPORT
One click downloads a zip: summary tables, figures and raw data.

LIVE VS STUDY AND SAVED RUNS



While recording: live numbers against the study (dashed line)





After the run: every test side by side, with a same-story check

SAVED RUNS (BOTTOM LEFT)
Each run kept with its packets, results and note; reopen or delete any time.

WHAT WE ASSUMED VS WHAT WE FOUND

We assumed	The case study found	Why
Losses are independent	Rejected: a loss makes the next 31 to 43× likelier ($p < 10^{-10}$).	Interference and full queues outlast the 20 ms between packets; the channel has memory.
Losses per second are Poisson	Rejected: variance ÷ mean 1.59 to 10.17; Negative Binomial wins.	Bursts pile losses into a few seconds and leave the rest empty.
Gaps between losses are exponential	Rejected (KS $p < 0.0005$); a two-phase mixture fits.	Two time scales: milliseconds inside a burst, seconds between bursts.
The usual 95% interval is fine	Held the truth only 40% to 86% of the time; corrected 93% to 94%.	Packets in a burst share one fate, so n ÷ deff is the real sample size.
All networks lose at the same rate	Rejected: Kruskal-Wallis H = 937, every pair differs.	Each link has different causes and amounts of loss.
A short capture is enough	Wired needs about 5 minutes for power 0.8.	Bursts are rare on a good link; power grows with length.

PHASE 2
The Lab's Assumptions tab adds a fourth column with what each real run shows.

DISCUSSION

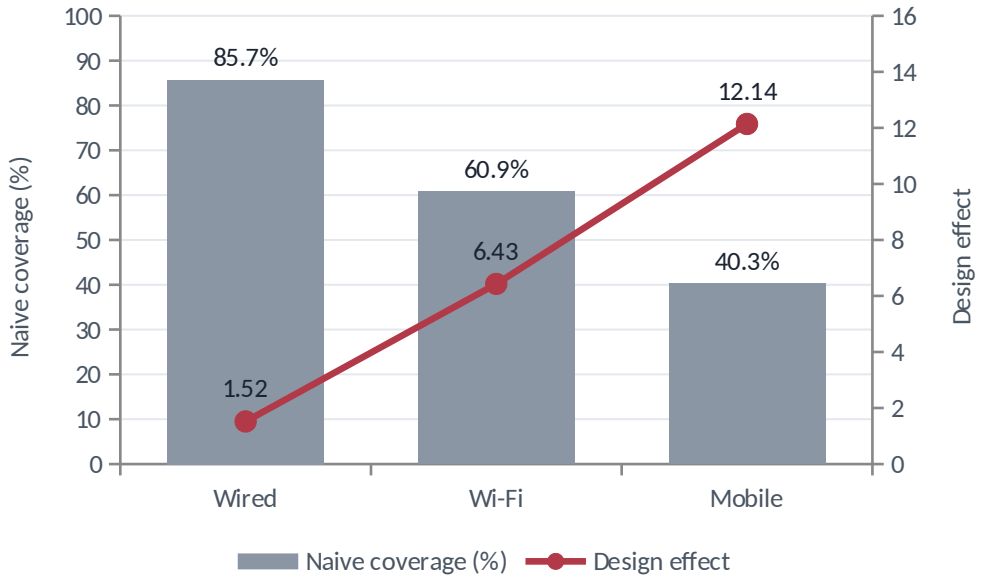
WHAT THE RESULTS MEAN

Burstiness is its own property. Mobile loses 14 times more than Wired. On both, a loss makes the next 31 to 43 times likelier.

Stronger bursts, worse intervals. Naive coverage is lowest on Mobile: 40.3% instead of 95%.

One step fixes it. Dividing n by the design effect gives 93.3% to 94.0% coverage.

NAIVE COVERAGE FALLS AS DESIGN EFFECT GROWS



WHY IT MATTERS IN PRACTICE

VIDEO AND VOICE
Error correction should cover a burst, not one packet. Mean burst: 1.12, 1.55 and 2.25 packets.

SERVICE REPORTS
A loss rate should come with a burst measure, such as P(L|L).

PROTOCOL TUNING
Retransmission timers tuned on random loss react badly to bursts.

STATISTICS
Check independence before trusting any standard error or interval.

LIMITATIONS

LIMITS OF THIS STUDY AND HOW WE HANDLE THEM

Limitation	Why it matters	How we handle it
Phase 1 data are simulated	Real links may add effects, such as busy hours.	Phase 2 lab runs the same code on real packets.
Dispersion test uses a Poisson baseline	Counts of 50 packets are Binomial, a little less spread out.	The test can only under-state burstiness, never over-state it.
Runs and Markov tests are approximate	In very short captures the false-alarm rate can drift from 5%.	Our traces last 1 hour, where the rate is close to 5%.
Tail models are not perfect	Negative Binomial and two-phase fits are still rejected for some networks.	Future work: hidden Markov and three-phase models.
Design effect depends on block size	Other block sizes give slightly different interval widths.	10-second blocks are far longer than any Bad spell.
One trace per network	A single run could be unusual.	The Monte Carlo studies repeat the experiment 2,000 times.

PACKET-LOSS LAB: SIMULATION PAGE

WHAT YOU CAN DO

Pick a network: Wired, Wi-Fi, Mobile.

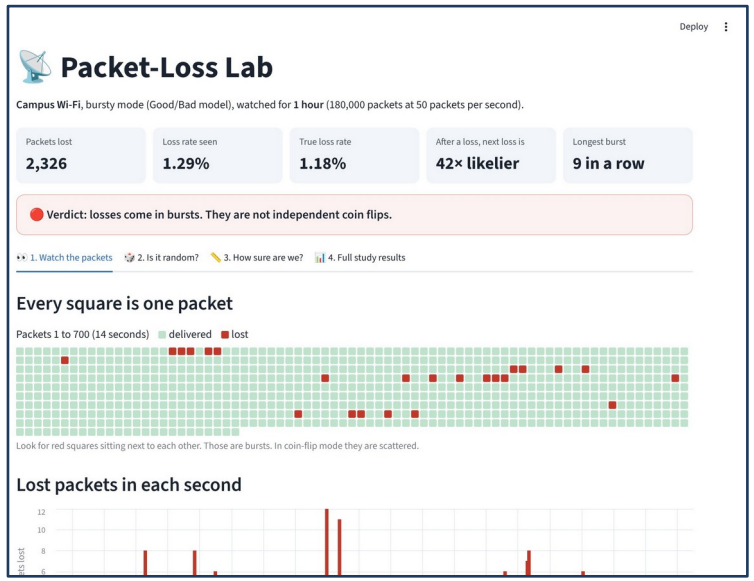
Coin-flip mode: same loss rate, but every packet independent.

Watch the packets: lost packets shown as red squares.

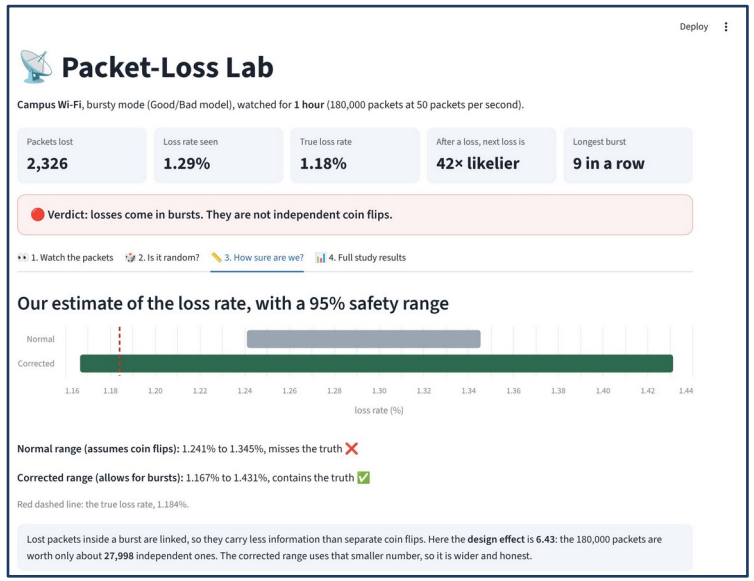
Is it random? the four tests, passed or failed.

How sure are we? naive and corrected intervals against the truth.

ONE-MINUTE DEMO
1. Wi-Fi: losses clump together. 2. Coin-flip mode: all checks pass. 3. Intervals: naive misses, corrected covers.



Watch the packets: red squares are losses



How sure are we: naive vs corrected

2,326
Wi-Fi losses, same as the study

42×
likelier after a loss

2026
default seed

RUN IT
streamlit run lab.py, then pick Simulation Lab (Phase 1) in the sidebar.

CONCLUSIONS AND FUTURE WORK

CONCLUSIONS

Loss counts: the Poisson model is rejected; the Negative Binomial has lower AIC.

Gaps: a fast phase (mean 29 to 33 ms) and a slow phase of seconds.

Dependence: P(L|L) is 31 to 43 times P(L|N).

Networks: mean loss rates of 0.28%, 1.29% and 3.94% differ significantly.

Intervals: naive coverage 40% to 86%; corrected 93% to 94%.

FUTURE WORK

Real runs: 10 minutes each on Wi-Fi, hotspot and wired with the Real Network Lab.

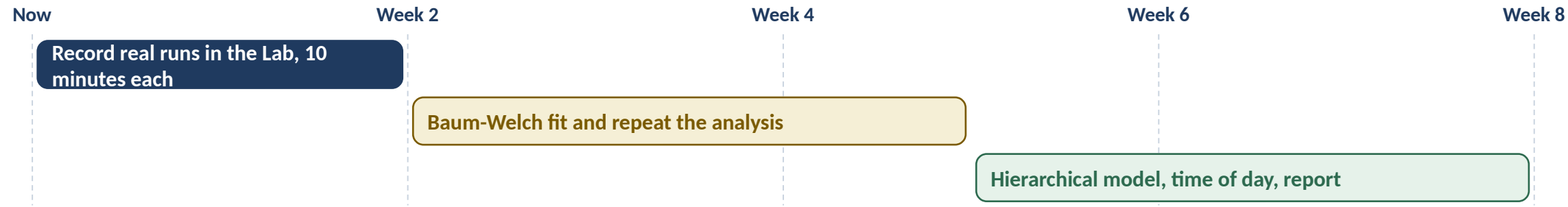
Calibration: estimate the two-state model with the Baum-Welch algorithm.

Tail models: three-phase mixtures or a hidden Markov model.

Hierarchical Bayes: pool across networks, with Phase 1 results as priors.

Time of day: compare peak and off-peak loss rates.

PROPOSED PHASE 2 TIMELINE (WEEKS AFTER THE FINAL REVIEW)



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[15] J. D. Hunter, "Matplotlib: a 2D graphics environment," Computing in Science and Engineering, vol. 9, no. 3, pp. 90-95, 2007.

USE OF SOURCES

Loss model [1, 2, 3]. EM algorithm [4]. Tests [5 to 9]. AIC [10]. Design effect [11]. Bayesian methods [12]. Software [13 to 15].

PAPER AND CODE

IEEE paper in the paper folder. Seed 2026 regenerates every number.

SOURCE CODE

github.com/AashmanBiyani08/Prob_Stats_Project

DATA

Phase 1 traces are simulated. Phase 2 measures real packets; validated on a test server.

APPENDIX: ABBREVIATIONS

MODELS AND FITTING

Short	Full form	Plain meaning
GE	Gilbert-Elliott model	Two-state Good/Bad channel that makes bursts
πB	Stationary Bad probability	Share of time spent in the Bad state
hG, hB	Loss probability in Good, Bad	Chance a packet is lost in each state
MLE	Maximum likelihood estimation	Picks values that make the data most likely
EM	Expectation-Maximisation	Guess, update, repeat: fits the two-phase mixture
AIC	Akaike Information Criterion	Model score; lower is better
NB	Negative Binomial	Count model that allows clumping
χ^2	Chi-square test	Compares observed and expected counts
KS	Kolmogorov-Smirnov test	Largest gap between data and model curves

TESTS AND INFERENCE

Short	Full form	Plain meaning
LRT, G^2	Likelihood-ratio test, its statistic	Does the previous packet matter?
P(L|N), P(L|L)	Loss after delivered, after lost	a and b; b/a is the burst amplification
H0, H1	Null, alternative hypothesis	No effect vs the effect we test for
α	Significance level	Reject H0 when $p < 0.05$
CrI	Credible interval	95% Bayesian range for the loss rate
deff	Design effect	Real spread ÷ spread under independence
n_eff	Effective sample size	n ÷ deff: worth of correlated packets
MC, SE	Monte Carlo, standard error	Repeat by simulation; its accuracy
pp	Percentage points	Difference between two percentages

APPENDIX: PHASE 2 TERMS

MEASURING

Term	Full form	Plain meaning
ICMP	Internet Control Message Protocol	The protocol behind ping; the router replies
UDP	User Datagram Protocol	Plain packets to our own echo server
RTT	Round-trip time	Time for a reply to come back (ms)
Echo server	Program that sends packets back	Used for UDP mode and the validation test
Timeout	2 s wait for a reply	No reply in time: the packet counts as lost
Router / gateway	First hop of the network	Probing it measures the Wi-Fi link itself

THE LAB

Term	Meaning	Plain meaning
Probe	probe.py	Sends the packets and records replies
Run / capture	One recording	CSV of packets plus a JSON of settings
Verdict	Bursty or no evidence	Runs test or Markov test at $\alpha = 0.05$
Live vs study	Side-by-side check	Real run against the matching study network
Same story	Same test outcome	Both reject, or both do not
Saved run	Kept run with results	Stored in realworld/saved_runs